(1.) Course Title: INFPHD644 Spatial Databases and Data Mining in Geoinformatics (complex exam preparatory)	**Credits: 6**
Course **Classification: Mandatory**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: ….. (credit%)	
Type of Class: practice and **number of hours**: **2** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: …**English**……)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: ……..	
Assessment Method (Exam / Midterm Grade / **Other**[3]): …….. **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: ……..	
Course Placement in Curriculum (Semester Number): …	
Prerequisites (*if any*): ……..	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

The exam focuses on the management, processing, and analysis of geospatial data in database and program
Students are expected to demonstrate theoretical understanding and practical skills in handling vector a
spatial databases (PostgreSQL with PostGIS), and applying geoinformatics data mining techniques. Stude
manage spatial data, including data types, spatial indexing, map projections, and topological concepts. Th
competence in using built-in geoprocessing functions, importing and exporting spatial data, and integra
web-based applications. The exam also covers geoinformatics data mining, including retrieving data fr
OpenStreetMap, processing vector and raster data using Python libraries (GDAL/OGR, Shapely, GeoPy,
GTFS and other geospatial data formats, converting between formats (e.g., text files to GeoJSON or KM
geospatial data. Students are required to integrate theoretical knowledge and practical skills to apply mo
tools effectively in research and applied contexts.

1. Fundamentals of database management: relational databases, records, fields, key fields, table relat
(B-tree, R-tree indexes), client–server systems, and storage engines. Describe a database managem
2. Structure and management of spatial data in a PostgreSQL/PostGIS environment. WKT and WKB
types. Spatial indexing. Handling map projections. Other functions that return geometric informati
3. Functions that return the dimensions of vector features. Distances, measurements, and enclosed an
4. Use of geometry-creating functions. Bounding geometries. Intersection, difference, symmetric difference, u
zones. Shortest line.
5. Geoprocessing tools (analysis of spatial relationships): intersects, crosses, overlaps, contains, etc.
6. Topological data structures (introduction to the topology module). Advantages of line simplification in a to
structure.
7. Handling raster data in databases in analytical workflows.
8. SQL in GIS software. Describe the possibilities for establishing database connections, performing database
importing and exporting vector and raster data to and from databases.
9. Data mining in geoinformatics. Describe the classical steps of data mining. Present the same process in geo
real-world problem.
10. Querying, cleaning, and processing geospatial data from various sources. Possible preprocessing steps usin
Python libraries (GDAL/OGR, Shapely, Pandas, GeoPandas, GeoPy, Matplotlib, etc.) for vector and raster

11. Transfer and transformation of spatial data between different formats (e.g., GeoJSON, KML, database files etc.).
12. Integration of spatial data into web applications and services.
13. Effective presentation of geospatial data for analysis, research, and decision-making. Information.
14. Introduction to the OpenStreetMap spatial database. Data structure, map features, and feature groups. Dow map data. Data quality.

2-5 Most Important Required and Recommended Literature

List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication de ISBN).

- Ungvári Zsuzsanna: Data mining in geoinformatics. Lecture notes. 2025. https://www.inf.elte.hu/dstore/document/227925/ungv%C3%A1ri_zsuzsanna_data%20mining.pdf
- Ungvári Zsuzsanna: Spatial Databases. Lecture notes. 2025. https://www.inf.elte.hu/dstore/document/227926/ungv%C3%A1ri_zsuzsanna_spatial_databases_j
- PostGIS Documentation: https://postgis.net/documentation/
- Nagy G.: Spatial databases online lecture notes 1-6. chapters.
  https://regi.tankonyvtar.hu/hu/tartalom/tamop425/0027_SDO1/index.html
  https://regi.tankonyvtar.hu/hu/tartalom/tamop425/0027_SDO2/index.html
  https://regi.tankonyvtar.hu/hu/tartalom/tamop425/0027_SDO3/index.html
  https://regi.tankonyvtar.hu/hu/tartalom/tamop425/0027_SDO4/index.html
  https://regi.tankonyvtar.hu/hu/tartalom/tamop425/0027_SDO5/index.html
  https://regi.tankonyvtar.hu/hu/tartalom/tamop425/0027_SDO6/index.html

- OpenStreetMap WIKI: https://wiki.openstreetmap.org/wiki/Main_Page, 2020
- Pang-Ning Tan, Michael Steinbach, Vipin Kumar: Introduction to Data Mining. Pearson Educatio 9780133128901, 0133128903. https://www-users.cs.umn.edu/~kumar001/dmbook/index.php#cha
- PostgreSQL Documentation: https://www.postgresql.org/docs/
- Ian H. Witten, Eibe Frank, Mark A. Hall, Christopher J. Pal: Data mining. Practical Machine Lear Techniques. ELSEVIER SCIENCE & TECHNOLOGY. 2017. ISBN 9780128042915 https://www.cs.waikato.ac.nz/~ml/weka/book.html
- Obe.E Regina: PostGIS in Action. Shelter Island, Manning, 2015. ISBN: 9781935182269
- Arsanjani, Zipf, Mooney, Helbich (eds.): OpenStreetMap in GIScience: Experiences, Research, ar Springer, 2015. ISBN: 9783319142807

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and preci boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the
- Possesses research methodology knowledge necessary for independent research in the given discipline/fi

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a nev conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new tec approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical back their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelatio personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation base characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable resea field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, an the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutio
- Participates in formulating theoretical and practical questions with a leadership role and high-level coope
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical profession.

Course responsible (name, position, academic degree): ……..……

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.

[2] e.g. case studies, role play, thematic presentations, etc.

[3] e.g. continuous debriefing, mid-year report

[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
Dr. Zuszsanna Ungvary, assistant professor, PhD